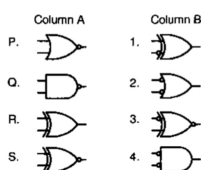


CBSE Mathematics Question Paper

Q.11 Match the logic gates in Column A with their equivalents in Column B.



Options:

- (A) P-2, Q-4, R-1, S-3
- (B) P-4, Q-2, R-1, S-3
- (C) P-2, Q-4, R-3, S-1
- (D) P-4, Q-2, R-3, S-1

Solution:

First identify each gate:

- **P:** OR gate with output bubble $\Rightarrow$ NOR gate
- **Q:** AND gate with output bubble $\Rightarrow$ NAND gate
- **R:** XOR gate
- **S:** XOR gate with output bubble $\Rightarrow$ XNOR gate

Now identify Column B:

- **1:** OR gate
- **2:** OR gate
- **3:** XOR with output bubble $\Rightarrow$ XNOR
- **4:** AND gate

Matching using logic equivalences:

- NOR is equivalent to OR with inverted inputs $\Rightarrow$ matches option 2
- NAND is equivalent to AND with inverted inputs $\Rightarrow$ matches option 4
- XOR matches XOR $\Rightarrow$ matches option 1
- XNOR matches XNOR $\Rightarrow$ matches option 3

$$P \rightarrow 2, \quad Q \rightarrow 4, \quad R \rightarrow 1, \quad S \rightarrow 3$$

Checking Options:

- (A) P-2, Q-4, R-1, S-3
Correct matching. **This is the right answer.**
- (B) P-4, Q-2, R-1, S-3
Incorrect: P (NOR) cannot match AND-type gate.
- (C) P-2, Q-4, R-3, S-1
Incorrect: R is XOR but matched with XNOR.
- (D) P-4, Q-2, R-3, S-1
Incorrect: Multiple mismatches.

Final Answer: $\boxed{(A)}$